

2.2 MATRIX, FIELD OF VIEW (FOV) & RESOLUTION

PIXEL SIZE, SPATIAL RESOLUTION & COVERAGE TRADE-OFFS

Matrix, FOV and slice thickness work together to determine spatial resolution, coverage and scan time. The technologist balances these factors to answer the clinical question efficiently.



KEY CONCEPT

- Resolution is about pixel (in-plane) and voxel (3D) size.
- Smaller pixels/voxels = more detail but require more data and longer scan time (or reduced SNR).

- ↑ Increase
- ↓ Decrease
- No / Minimal Effect

1 THE BASICS

FOV (Field of View)

The physical size of the area being imaged.



Larger FOV → more coverage, larger pixels.

MATRIX

Number of pixels in the phase (Freq) × frequency (Freq) directions.



Higher matrix → smaller pixels.

SLICE THICKNESS

The thickness of each slice (and gap between slices if any).



Thinner slices → better detail, longer scan.

2 IN-PLANE RESOLUTION (PIXEL SIZE)

$$\text{Pixel Size (mm)} = \frac{\text{FOV (mm)}}{\text{Matrix}}$$

Example (Frequency × Phase = 256 × 256)

FOV (mm)	Matrix	Pixel Size (mm)
220	256	0.86
200	256	0.78
180	256	0.70
160	256	0.63

↑ Increase FOV → Larger pixels (less detail)

↓ Increase Matrix → Smaller pixels (more detail)

3 VOXEL SIZE (3D RESOLUTION)

$$\text{Voxel Size} = \text{Pixel Size} \times \text{Slice Thickness}$$

Example (FOV 220 mm, Matrix 256)

Pixel Size	Slice Thickness	Voxel Size (approx.)
0.86 mm	5.0 mm	$0.86 \times 5.0 = 4.30 \text{ mm}^3$
0.86 mm	3.0 mm	$0.86 \times 3.0 = 2.58 \text{ mm}^3$
0.86 mm	1.5 mm	$0.86 \times 1.5 = 1.29 \text{ mm}^3$
0.86 mm	1.0 mm	$0.86 \times 1.0 = 0.86 \text{ mm}^3$

↑ Thinner slices → Smaller voxels (better 3D detail)

↓ Thinner slices → Lower SNR, longer scan

4 HOW PARAMETERS AFFECT EACH OTHER

Parameter	Increase	Decrease	Effect on...
FOV	Larger coverage Bigger pixels	Smaller coverage Smaller pixels	Detail ↓ SNR ↑ Scan Time ↑
Matrix	Smaller pixels	Larger pixels	Detail ↑ SNR ↓ Scan Time ↑
Slice Thickness	Lower SNR Longer scan	Higher SNR Shorter scan	Detail (Z) ↑ SNR ↓ Scan Time ↓

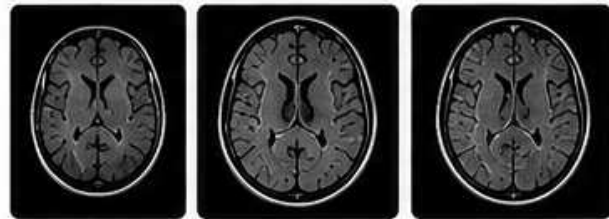
5 VISUAL EXAMPLES – SAME ANATOMY

A. FOV (Matrix 256)

FOV 160 mm

FOV 220 mm

FOV 320 mm



Smaller FOV = more detail

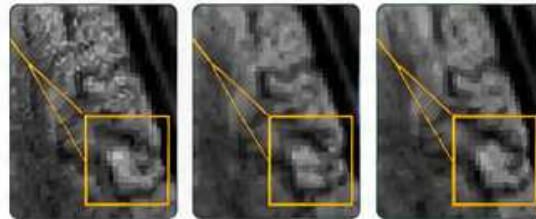
Larger FOV = more coverage

B. MATRIX (FOV 220 mm)

Matrix 128

Matrix 192

Matrix 256



Lower Matrix = blocky

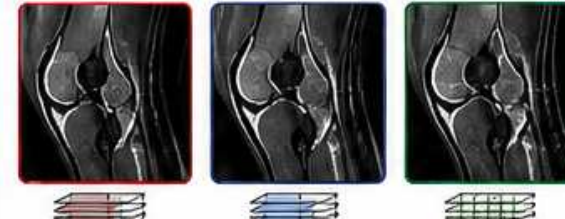
Higher Matrix = sharper

C. SLICE THICKNESS (FOV 220 mm, Matrix 256)

5.0 mm

2.0 mm

1.0 mm



Thicker = less detail

Thinner = more detail

6 COVERAGE EXAMPLE (AXIAL BRAIN)

Same prescription, different FOV

FOV 160 mm

FOV 240 mm



Less coverage (cuts anatomy)



More coverage (includes anatomy)

7 SCAN TIME & SNR CONSIDERATIONS

Scan Time Increases When You:

- Increase matrix
- Decrease slice thickness
- Increase number of slices (coverage)
- Use higher bandwidth (usually decreases time slightly)

SNR Decreases When You:

- Increase matrix
- Decrease slice thickness
- Decrease NEX (averages)
- Use very large FOV



Increasing scan time does NOT inherently lower SNR. Deliberately increasing scan time (e.g., by increasing NEX) is a common way to improve SNR. $\text{SNR} \propto \sqrt{\text{NEX}}$ (doubling NEX → ~1.4× SNR)

8 PRACTICAL GUIDELINES

Goal	What to do
Highest spatial detail (research)	Small FOV, high matrix, thin slices, accept longer scan / lower SNR
Routine clinical exam	Moderate FOV, appropriate matrix, slices matched to anatomy
Fast exam / patient comfort	Larger FOV, lower matrix, thicker slices, higher parallel imaging
Large anatomy (spine, body)	Increase FOV and number of slices to ensure full coverage



There is always a trade-off between resolution, SNR, coverage and scan time. Optimize based on the clinical question.

9 ENGINEER CHECKLIST

- ✓ Is the anatomy fully covered in all three dimensions?
- ✓ Is pixel size appropriate for the clinical question?
- ✓ Is slice thickness appropriate (consider partial volume)?
- ✓ Are scan time and SNR acceptable for the protocol?
- ✓ Could a different combination achieve the same goal faster?



KEY TAKEAWAY

Matrix, FOV and slice thickness determine the size of the voxel and the amount of detail in the image.

There is always a trade-off between resolution, SNR, coverage and scan time.

Choose the right balance for the right clinical question.



RIGHT INFORMATION. RIGHT BALANCE. RIGHT IMAGE. Use the parameters. Understand the trade-offs. Deliver the value.